

Mohit Bajaj

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Experience

Cuemath

Jul 2024 – Apr 2026

Software Development Engineer

- Built real-time multiplayer learning games architecture using React and WebSockets (Pub/Sub), enabling synchronized gameplay between multiple users
- Designed and implemented core gameplay systems including question flow, timers, scoring, leaderboards, and lives/energy mechanics
- Developed reusable game architecture with shared utilities and base classes, reducing time required to build new games (responsive grids, animation players, containerized physics elements e.tc.)
- Handled real-time state synchronization across players, ensuring consistent game state and low-latency interactions
- Integrated gameplay systems across web and mobile platforms using React and React Native

Cuemath

Nov 2023 – Jul 2024

SDE Intern

- Developed browser-based interactive games using Phaser and integrated them with React applications
- Built reusable game utilities and modular components to support rapid development of multiple game types
- Implemented real-time updates using WebSockets to support live gameplay interactions

Frugalx

May 2023 – Jul 2023

Web-ML Intern

- Built web interfaces integrating machine learning outputs into interactive user features

Projects

Browser-Based Operating System | *Next.js, Zustand, React, TypeScript*

- Built a fully interactive operating system inside the browser with window management, file system, and multi-app architecture
- Implemented shared state system enabling cross-application file access and persistence across sessions
- Designed macOS-inspired UI with dynamic windowing, dock system, and app lifecycle management

3D Avatar Streaming System | *React, Three.js, WebRTC, PeerJS*

- Developed real-time 3D avatar streaming system enabling low-latency interaction between users
- Integrated WebRTC-based communication pipelines for audio, messaging, and avatar synchronization
- Built customization system using Mediapipe-driven inputs for interactive user control

3D Interactive Portfolio | *React, Three.js*

- Built interactive portfolio with custom 3D rendering, animations, and dynamic UI transitions
- Integrated mini-game experiences and audio interactions to enhance user engagement

Technical Skills

Languages: JavaScript, TypeScript, Python, C++

Game & Graphics: Phaser, Three.js, Canvas APIs

Frameworks: React, React Native, Next.js, Node.js

Game Systems & Concepts: Game State Management, Real-time Multiplayer Systems, Event-driven Architecture, Gameplay Logic (timers, scoring, leaderboards)

Networking: WebSockets (Pub/Sub), WebRTC

Other: Data Structures & Algorithms, System Design, Performance Optimization, Git, AI-assisted development workflows

Education

Rajasthan Technical University

2024

B.E. / B.Tech in Computer Science

- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems